

Research Gaps, Objectives, and Scope

Project: Evaluating and Improving Cross-Dataset Generalization of ML-Based Intrusion Detection Systems **Document:** 03 — Research Gap, Objectives & Scope **Status:** Prepared for mid-semester review

1. Problem Statement (one paragraph)

Machine-learning-based Intrusion Detection Systems (IDS) are routinely reported with near-perfect accuracy — but almost always when *trained and tested on the same dataset*. Recent work [1]–[4] shows that a model trained on one network-traffic dataset and deployed on a *different* one often performs no better than random guessing, because it has learned dataset-specific artefacts (identity features, ports, collection-window statistics, protocol quirks) rather than genuine attack behaviour. At the same time, a mature body of work uses bio-inspired/metaheuristic optimisation (GA, PSO, NSGA-II/III, etc.) to select features for IDS [8]–[12] — but **always with a fitness function that maximises single-dataset accuracy**. This project connects the two: we use metaheuristic optimisation to search for **features and models that remain accurate across datasets**, and we benchmark whether doing so narrows the generalisation gap.

2. Research Gaps (formal)

Derived from the literature review (02_literature_review.md), three gaps exist:

- **Gap 1 — Optimisation-for-transfer is unexplored.** No published work performs metaheuristic/bio-inspired *feature selection* whose fitness is **cross-dataset** generalisation. Existing metaheuristic FS optimises same-dataset accuracy [8]–[12]; existing cross-dataset studies are diagnostic [1]–[3] or model-centric domain adaptation that needs target-domain data [13].
- **Gap 2 — No shared benchmark across the three heterogeneous datasets.** There is no systematic study that (i) harmonises **CIC-IDS-2017, UNSW-NB15, and TON_IoT** onto a common feature subspace, (ii) evaluates the *same* models cross-dataset, and (iii) reports *which features and attack classes transfer* versus which are collection artefacts.
- **Gap 3 — No consolidated feature-level guidance.** No work translates the generalisation evidence into practical, feature-level guidelines for building IDS that transfer to unseen network environments.

3. Novelty / Contribution

1. **A transfer-oriented optimisation objective.** We define and implement a metaheuristic feature-selection fitness that rewards *cross-dataset* robustness (average F1/balanced accuracy across source→target pairs), not same-dataset accuracy. To our knowledge this is the first such objective in the NIDS literature.
2. **A reproducible 3-dataset cross-dataset benchmark** over a documented shared feature subspace (see 01_dataset_report.md §6.2), with leakage features removed and macro metrics reported under realistic class distributions (addressing the protocol pitfalls in [3]).
3. **Characterisation of the generalisation gap**, identifying transferable vs dataset-specific features and attack classes (fills Gap 2).

4. **Comparison of multiple bio-inspired optimisers** (GA, PSO, and at least one more — Differential Evolution or Whale Optimization) against standard feature-selection baselines (mutual information, RF importance, PCA, RFE, LASSO).
5. **Practical guidelines** distilled from the results (fills Gap 3).

4. Objectives

The five objectives below refine and operationalise the project proposal. Each maps to a deliverable in 04_methodology_and_plan.md.

1. **Evaluate cross-dataset generalisation of multiple ML/DL models** — train Random Forest, XGBoost/LightGBM, SVM, Logistic Regression, and a small MLP on one dataset and test on the structurally different ones (CIC-IDS-2017, UNSW-NB15, TON_IoT), in addition to same-dataset splits. (*Metrics: per-class F1, balanced accuracy, macro-F1; full 3×3 source→target matrix.*)
2. **Characterise the sources of the generalisation gap** — determine which attack categories, features, and traffic patterns hold up across environments vs which are dataset-specific artefacts (identity features, ports, collection-window ct_* features). (*Methods: feature-importance/SHAP analysis, t-SNE/UMAP visualisation, per-class confusion.*)
3. **Explore and compare metaheuristic/bio-inspired optimisers for transfer-oriented feature selection and model tuning** — Genetic Algorithm, Particle Swarm Optimization, and additional swarm/evolutionary methods (e.g. Differential Evolution, Whale Optimization), with fitness = cross-dataset robustness.
4. **Benchmark against standard feature-selection and tuning baselines** — mutual information, random-forest importance, PCA, recursive feature elimination, LASSO, and default hyper-parameters — to quantify how much each technique narrows the gap.
5. **Consolidate findings into practical guidelines** for building IDS that generalise reliably to previously unseen network environments.

5. Scope and Design Decisions

These decisions were fixed after a first analysis of the data and literature (see Dataset Report §6–7); they bound the work to keep it tractable for a two-person BTech project.

Decision	Choice	Rationale
Classification granularity	Binary (benign vs attack) as primary; coarse multi-class (~6 classes) as secondary	Binary is the dominant generalisation framing [1], [2]; multi-class class vocabularies barely overlap across datasets (Dataset Report §6.3)
Model family	Classic ML (RF, XGBoost/LGBM, SVM, LR) + one small MLP	Fast enough to run <i>inside</i> a GA/PSO fitness loop; interpretable for feature characterisation
Optimisation target	Feature subset (and light-weight hyper-parameter tuning)	Feature selection is where the transfer signal lives; it is cheap to evaluate
Feature space	Shared semantic subspace (~15–25 dims) for cross-dataset runs; full features only for same-dataset baselines	Schema mismatch makes full-feature transfer impossible (Dataset Report §6)
Leakage handling	Strip IP/port/ID/time-stamp features	Otherwise models memorise identity, not behaviour [1], [3]
Metrics	Balanced accuracy, macro-F1, per-class F1 (not overall accuracy)	Class imbalance is dataset-specific and severe in CICIDS2017
Evaluation protocol	Train-on-source / test-on-target, fixed seeds, repeated runs	Reproducibility; avoids the protocol pitfalls of [3]

6. Expected Outcomes

1. A quantitative answer to “*how bad is the generalisation gap across CIC-IDS-2017, UNSW-NB15, TON_IoT?*” (baseline matrix).
2. Evidence on “*does transfer-oriented metaheuristic feature selection beat standard feature selection at narrowing that gap?*” (the core claim).
3. A ranked list of *transferable vs artefact* features and a per-attack-class transfer map.
4. A short set of actionable guidelines for building generalisable IDS.

7. Risks and Mitigations

Risk	Mitigation
Shared subspace too small $\rightarrow$ weak models	Expand subspace with derived features (rates, ratios); report subspace size sensitivity
GA/PSO fitness too slow (MLP inside loop)	Use fast inner models (RF/XGB) for FS; reserve MLP for post-selection evaluation
Cross-dataset scores near random (per [1])	Frame contribution as <i>characterisation + relative improvement over baselines</i> , not absolute accuracy
Memory/time on large CSVs	Downsample with stratification; cache harmonised matrices as .parquet

8. References

Reference numbers follow the canonical 15-paper list in 02_literature_review.md.

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- [2] M. Z. Hossain et al., “Assessing generalisation capability of machine learning models for intrusion detection,” arXiv:2605.04407, 2026.
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- [4] M. U. Butt, A. Hotho, and D. Schlör, “Evaluating tabular representation learning for network intrusion detection,” in *Proc. IEEE CSR*, 2026.
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- [7] N. Moustafa, “A new distributed architecture for evaluating AI-based security systems at the edge: Network TON_IoT datasets,” *Sustain. Cities Soc.*, vol. 72, 2021.
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- [15] J. Vitorino, M. Silva, E. Maia, and I. Praça, “An adversarial robustness benchmark for enterprise network intrusion detection,” in *Proc. FPS*, 2023.